

Week 5: Unsupervised Learning - Pre-Class Study Guide

Time Required: 30 minutes **Purpose:** Prime your understanding of unsupervised learning before the live session

Welcome to Week 5!

You've mastered **supervised learning** (Weeks 1-4) - predicting targets when you have labels. This week we're tackling **unsupervised learning** - finding patterns when you DON'T have labels!

The shift: No more “y” column. No ground truth. The algorithm discovers hidden structure in data.

Real-world examples: - Customer segmentation (no pre-labeled customer types) - Document clustering (group similar articles automatically) - Dimensionality reduction (visualize 100D data in 2D)

Pre-Class Activities (Complete Before Live Session)

Part 1: Watch These Videos (26 minutes total)

Watch in this order:

1. **StatQuest - K-Means Clustering** (9 min) <https://www.youtube.com/watch?v=4b5d3muPQmA>

What to focus on:

- The 4 steps: Initialize → Assign → Update → Repeat
- How centroids “move” toward cluster centers
- Why it's called **K-Means**

2. **StatQuest - PCA Main Ideas in Only 5 Minutes** (5 min) https://www.youtube.com/watch?v=HMOI_lkzW08

What to focus on:

- PCA finds “directions of maximum variance”
- Dimensionality reduction: 100 features → 2 features
- Why we need this (can't visualize 100 dimensions!)

3. **StatQuest - PCA Step-by-Step** (12 min) <https://www.youtube.com/watch?v=FgakZw6K1QQ>

What to focus on:

- How PCA rotates data
- Variance explained ratio

- Scree plots for choosing components
 - You DON'T need to understand eigenvectors mathematically!
4. **Numiqo – PCA** (18 mins) https://www.youtube.com/watch?v=__6UjscCJrYE&t=228s
- Interactive** <https://numiqo.com/lab/pca>
- What to focus on:** – The same as seen in the above videos just presented differently (It is really good!) —

Part 2: Key Concepts to Understand

By the end of the videos, you should have a basic understanding of:

Supervised vs Unsupervised Learning

Supervised Learning	Unsupervised Learning
Training data: X + y (features + labels)	Training data: X only (features, no labels)
Goal: Predict y from X	Goal: Find patterns/structure in X
Examples: Regression, Classification	Examples: Clustering, Dimensionality Reduction
Evaluation: Compare to true labels	Evaluation: No ground truth! Use quality metrics
Weeks 1-4	Week 5

When to Use Unsupervised Learning

Scenario 1: No Labels Available

- Customer segmentation (no pre-labeled segments)
- Discovering news article topics
- Why: Creating labels is expensive or impossible

Scenario 2: Pattern Discovery

- Anomaly detection (unusual network traffic)
- Market basket analysis (what products are bought together?)
- Why: Don't know what patterns exist beforehand

Scenario 3: Dimensionality Reduction

- Visualize high-dimensional data (1000 features $\rightarrow$ 2D plot)
- Preprocessing for supervised learning (reduce noise)
- Why: Can't visualize or compute efficiently with too many features

K-Means Clustering (Preview)

What it does: Groups similar data points into K clusters

The algorithm (intuitive):

1. **Initialize:** Place K random cluster centers (centroids)
2. **Assign:** Each point joins its nearest centroid's cluster
3. **Update:** Move centroids to the mean of their cluster members
4. **Repeat:** Steps 2-3 until centroids stop moving (convergence)

Key questions we'll answer:

- How do we choose K (number of clusters)?
- How do we evaluate if clustering is good? (No labels to compare!)
- What makes K-Means different from classification?

This week's example: Clustering mall customers by income and spending habits to discover customer segments (Budget Shoppers, High Rollers, etc.)

PCA - Principal Component Analysis (Preview)

What it does: Reduces dimensions while preserving information

Intuition:

- Imagine 784 features (MNIST: 28x28 pixel images)
- PCA finds the “best viewing angle” - directions with most variance
- Projects data onto these directions (principal components)
- Result: 784 dimensions $\rightarrow$ 2 dimensions (for visualization!)

Key concepts:

- **PC1 (Principal Component 1):** Direction with MOST variance
- **PC2:** Second most variance (perpendicular to PC1)
- **Variance explained:** How much information each component captures
- **Scree plot:** Helps choose how many components to keep

This week's example: Visualizing MNIST handwritten digits (64D $\rightarrow$ 2D scatter plot where similar digits cluster together!)

Part 3: Vocabulary Pre-Reading

These terms will appear frequently in Week 5. You don't need to memorize them now - just get familiar:

Term	Quick Definition
Unsupervised Learning	ML when training data has features (X) but no labels (y)
Clustering	Grouping similar data points together
Centroid	Center of a cluster (mean position of all cluster members)
K-Means	Clustering algorithm that finds K cluster centers iteratively
Inertia (WCSS)	Sum of squared distances from points to their centroid (lower = tighter clusters)
Elbow Method	Plot inertia vs K, look for “elbow” to choose optimal K
Silhouette Score	Metric for cluster quality (range: -1 to 1, higher = better separation)
Dimensionality Reduction	Reducing number of features while preserving information
PCA	Principal Component Analysis - finds directions of maximum variance
Principal Components (PC)	New axes found by PCA, ordered by variance explained
Variance Explained	Percentage of total variance captured by each component
Scree Plot	Plot of variance vs component number, shows how many components to keep
Feature Scaling	Standardizing features to mean=0, std=1 (CRITICAL for K-Means and PCA!)

Part 4: Connection to Weeks 1-4

What stays the same:

- sklearn API pattern (create model → fit → predict/transform)
- Feature scaling with **StandardScaler**
- Data exploration and visualization
- Train/test splits (for supervised tasks AFTER clustering)

What's new:

- No labels (y) during training
- Can't evaluate with accuracy/RMSE (no ground truth!)
- Interpretation is subjective (“What does Cluster 2 represent?”)
- Different sklearn methods:
 - **fit()** still trains the model
 - **predict()** assigns clusters (K-Means)
 - **transform()** projects data (PCA)

The pattern you know still works!

```
# Week 1-4: Supervised
model = LogisticRegression()
model.fit(X_train, y_train)           # Train with labels
y_pred = model.predict(X_test)

# Week 5: Unsupervised
model = KMeans(n_clusters=3)
model.fit(X)                          # Train WITHOUT labels!
clusters = model.predict(X)          # Assign cluster labels
```

Part 5: Questions to Think About

Come to the live session with thoughts on these questions:

1. **Customer Segmentation:** A retail company has 10,000 customers with data on age, income, and purchase frequency. They want to create targeted marketing campaigns. Why is this an unsupervised learning problem? (Hint: Do they have pre-labeled customer types?)
2. **Choosing K:** If you're clustering customers, would you prefer K=2 (two segments) or K=100 (one hundred segments)? What are the tradeoffs?
3. **Interpretation:** K-Means assigns customer 1234 to Cluster 3. What does "Cluster 3" mean? How would you explain it to a marketing manager who doesn't know ML?
4. **PCA for Visualization:** You have a dataset with 50 features. You want to visualize it in a 2D scatter plot. Why can't you just pick 2 random features? What does PCA offer?

These questions have nuanced answers - we'll explore them during the live session!

Part 6: Pre-Class Practice (Optional but Recommended)

If you have an extra 10-15 minutes, try this quick mental exercise:

Clustering Exercise:

Imagine clustering these 9 customers by income and spending:

Customer	Income	Spending
A	Low	Low
B	Low	Low
C	Low	High

D	High	Low
E	High	Low
F	High	High
G	Medium	Medium
H	High	High
I	Low	High

Questions:

1. If $K=2$, how might K-Means group them?
2. If $K=3$, what three segments might emerge?
3. Give each cluster a business-friendly name (e.g., “Budget Shoppers”)

Possible $K=3$ solution:

- Cluster 0: A, B (Low/Low) → “Budget Shoppers”
- Cluster 1: D, E (High/Low) → “Conservative Wealthy”
- Cluster 2: C, F, H, I (High spending) → “Big Spenders”
- G: Could go to any cluster (it’s in the middle!)

Note: There’s no “right answer”! K-Means might group differently. That’s the nature of unsupervised learning!

What to Expect in the Live Session

Schedule:

- 0:00-0:15: Week 4 recap & Unsupervised vs Supervised
- 0:15-0:35: Customer segmentation metaphor (the guiding example!)
- 0:35-1:10: K-Means deep dive (algorithm, elbow method, silhouette)
- 1:10-1:25: **BREAK**
- 1:25-1:55: PCA theory (variance explained, scree plots)
- 1:55-2:15: Live coding - Mall Customers K-Means clustering
- 2:15-2:35: Live coding - MNIST PCA visualization (stunning!)
- 2:35-2:55: Pair programming - “Cluster Detective” exercise
- 2:55-3:00: Wrap-up & Week 6 preview

What to bring:

- Questions from the videos
 - Your Week 4 homework (we’ll discuss validation methodology)
 - Notebook for taking notes
 - Curiosity about pattern discovery!
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Success Checklist

Before the live session, you should:

- ☐ Watched StatQuest K-Means video (9 min)
 - ☐ Watched StatQuest PCA videos (5 min + 12 min)
 - ☐ Reviewed vocabulary list
 - ☐ Thought about the discussion questions
 - ☐ (Optional) Tried the mental clustering exercise
 - ☐ Ready to explore unsupervised learning!
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If You're Feeling Overwhelmed

Don't worry! Unsupervised learning feels abstract at first because there's no "correct answer" to check against. The live session is designed to make everything concrete through:

- Customer segmentation example (real business application)
- Visual demonstrations (see clusters form!)
- Live coding with Mall Customers data
- Stunning MNIST visualization (you'll SEE why PCA works)
- Pair programming practice

Remember: You don't need to master these concepts before class. The pre-work primes your brain. The real learning happens during the live session with hands-on practice!

The mindset shift: In supervised learning, we ask "Is my prediction correct?" In unsupervised learning, we ask "Do these patterns make sense for my problem?" This feels uncomfortable at first - that's normal!

Questions Before Class?

If anything is unclear, don't stress! The live session will bring it all together.

Common questions we'll answer:

- "How is clustering different from classification?"
 - "How do I know if my clusters are good with no labels?"
 - "What's the 'right' number of clusters?"
 - "Can I use PCA before supervised learning?"
 - "Why do we need to scale features for K-Means and PCA?"
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Week 5 Datasets (Preview)

Mall Customers (Pre-class, Live session K-Means, Pair programming)

- 200 customers

- Features: CustomerID, Gender, Age, Annual Income, Spending Score
- Goal: Discover customer segments for marketing

MNIST Digits (Live session PCA)

- 1,797 handwritten digits (0-9)
- 64 features (8x8 pixel images)
- Goal: Visualize high-dimensional data in 2D

Wine (Post-class homework)

- 178 wine samples
- 13 chemical features (alcohol, acidity, sugar, etc.)
- Goal: Cluster wines by chemical properties, interpret segments

Iris (Bonus exercise - PCA + supervised pipeline)

- 150 flowers, 4 features
- Goal: Use PCA for preprocessing before classification

See you in the live session!

Week 5 Pre-Class Study Guide v1.0 / January 2026